Bin Cheng (程斌)

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Google Scholar
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Research Interest

My research interest lies in **quantum information science**, with a particular focus on

- Quantum algorithms in the NISQ/early fault-tolerant regime
- Quantum error correction
- Quantum computation and combinatorial structures
- Classical simulation of quantum systems
- Quantum advantage
- Quantum complexity theory

Employment & Education

2023–now *Postdoctoral Research Fellow*
Centre for Quantum Technologies, National University of Singapore
Hosted by Patrick Rebentrost

2019–2024 *Ph.D. in Computer Science*
University of Technology Sydney (UTS)
Advisors: Zhengfeng Ji, Michael Bremner and Man-Hong Yung (external)
Ph.D. Thesis: *Classical Verification and Enhancement of Near-Term Quantum Devices*

2014–2018 *B.Sci. in Physics*
Southern University of Science and Technology (SUSTech)
Advisor: Man-Hong Yung

Publications & Preprints

1. Bin Cheng, Naixu Guo, Feng Pan
   Certifying Quantum Code and Circuit Distances at Scale. arXiv:2611.08247
2. Bujiao Wu, Tongyin Lin, Bin Cheng (corresponding author)
   Substantially Improved Rate and Distance for Addressable Transversal Phase Gates. arXiv:2611.04964
3. Daniel J. Spencer, Bin Cheng, Dinh-Long Vu, Kishor Bharti, Alexey V. Gorshkov, Patrick Rebentrost
   Pricing of graph-based quantum portfolios from first principles. arXiv:2610.03279
4. Bin Cheng, Feng Pan
   Exact Maximum Likelihood Decoding beyond Treewidth via Rank-Decomposition Dynamic Programming. arXiv:2609.39556
5. Tongyin Lin, Bujiao Wu, Bin Cheng (corresponding author)
   Asymptotically Good Quantum Codes with Addressable Transversal T Gates. arXiv:2609.35129
6. Fei Meng*, Bin Cheng*, Jianan Li, Man-Hong Yung (*: equal contribution, corresponding author)
   Weak Permanent Anti-Concentration for Random Gaussian Matrices in Boson Sampling. QCE 26, Toronto. arXiv:2607.22088
7. Jeong Yu Han, Bin Cheng, Dinh-Long Vu, Patrick Rebentrost. (corresponding author)
   Quantum Advantage for Multi-option Portfolio Pricing and Valuation Adjustments. Quantitative Finance (2026)
8. Zhenyu Chen*, Bin Cheng*, Minbo Gao, Xiaodie Lin, Ruiqi Zhang, Zhaohui Wei, and Zhengfeng Ji (*: equal contribution)
   Scalable Quantum Error Mitigation with Neighbor-Informed Learning. arXiv:2512.12578
9. Bin Cheng, Ziyuan Wang, Ruixuan Deng, Jianxin Chen, and Zhengfeng Ji
   Breaking the Treewidth Barrier in Quantum Circuit Simulation with Decision Diagrams. QIP 2026, arXiv:2510.06775
10. Dinh-Long Vu, Bin Cheng, Patrick Rebentrost (corresponding author)
    Low Depth Amplitude Estimation without Really Trying. ACM Transactions on Quantum Computing (2025)

11. Ziyuan Wang*, Bin Cheng*, Longxiang Yuan, Zhengfeng Ji (*: equal contribution)
FeynmanDD: Quantum Circuit Analysis with Classical Decision Diagrams. In Computer Aided Verification (2025)

12. Michael J. Bremner, Bin Cheng, Zhengfeng Ji (corresponding author, alphabetical order)
Instantaneous Quantum Polynomial-time Sampling and Verifiable Quantum Advantage: Stabilizer Constructions and Classical Security. PRX Quantum (2025)

13. Bin Cheng*, Fei Meng*, Zhi-Jiong Zhang, Man-Hong Yung (*: equal contribution)
Generalized Cross-Entropy Benchmarking for Random Circuits with Ergodicity. Chip (2025)

14. Patrick Rebentrost, Alessandro Luongo, Bin Cheng, Samuel Bosch, Seth Lloyd (corresponding author)
Quantum Computational Finance for Martingale Asset Pricing in Incomplete Markets. Scientific Reports (2024)

15. Xiao-Wei Li, Xiao-Ming Zhang, Bin Cheng, Man-Hong Yung
Reachability Deficit of Variational Grover Search. Physical Review A (2024)

16. Bin Cheng, Xiu-Hao Deng, Xiu Gu, *et al.*
Noisy Intermediate-scale Quantum Computers. Frontiers of Physics (2023)

17. Chong Ying*, Bin Cheng*, Youwei Zhao*, He-Liang Huang, Yu-Ning Zhang, Ming Gong, Yulin Wu, Shiyu Wang, Futian Liang, Jin Lin, Yu Xu, Hui Deng, Hao Rong, Cheng-Zhi Peng, Man-Hong Yung, Xiaobo Zhu, Jian-Wei Pan (*: equal contribution)
Experimental Simulation of Larger Quantum Circuits with Fewer Superconducting Qubits. Physical Review Letters (2023)

18. Fan Wang*, Bin Cheng*, Zi-Wei Cui, Man-Hong Yung (*: equal contribution)
Quantum Computing by Quantum Walk on Quantum Slide. arXiv:2211.08659

19. Kai Sun, Zi-Jian Zhang, Fei Meng, Bin Cheng, Zhu Cao, Jin-Shi Xu, Man-Hong Yung, Chuan-Feng Li, Guang-Can Guo
Experimental Verification of Group Non-membership in Optical Circuits. Photonics Research (2021)

20. Xi Chen*, Bin Cheng*, Zhaokai Li, Xinfang Nie, Nengkun Yu, Man-Hong Yung, Xinhua Peng (*: equal contribution)
Experimental Cryptographic Verification for Near-Term Quantum Cloud Computing. Science Bulletin (2021)

21. Bujiao Wu*, Bin Cheng*, Fei Jia, Jialin Zhang, Man-Hong Yung, Xiaoming Sun (*: equal contribution)
Speedup in Classical Simulation of Gaussian Boson Sampling. Science Bulletin (2020)
22. Man-Hong Yung, Bin Cheng (corresponding author)
Anti-Forging Quantum Data: Cryptographic Verification of Quantum Cloud Computing. arXiv:2005.01510
23. Bin Cheng, Man-Hong Yung.
Ubiquitous Complexity of Entanglement Spectra. arXiv:1906.00324

Talks

- **Weak Permanent Anti-Concentration for Random Gaussian Matrices in Boson Sampling**
IEEE QCE 2026, in Toronto, Canada.
- **Breaking the Treewidth Barrier in Quantum Circuit Simulation with Decision Diagrams**
29th Annual Quantum Information Processing Conference (QIP) 2026, in Riga, Latvia.
- **FeynmanDD: Quantum Circuit Analysis with Classical Decision Diagrams**
Institute of Physics Singapore (IPS) meeting 2025, in Singapore.
- **Instantaneous Quantum Polynomial-time Sampling and Verifiable Quantum Advantage: Stabilizer Constructions and Classical Security**
Joint CQT-Munich Center for Quantum Science and Technology (MCQST) Workshop 2025, in Singapore.
Joint CS group meeting at CQT, 2024.
CQT CS Seminar, 2023.
- **Experimental Cryptographic Verification for Near-Term Quantum Cloud Computing**
Nagoya-SUSTech Quantum Information Workshop 2019, in Nagoya, Japan.
Mini-Workshop on Quantum Verification 2019, in Shanghai, China.

Grants

- CQT Young Researcher Career Development Grant, Centre for Quantum Technologies, 2026–2028 (S$15,000)
  Project title: *Fundamental Limits of Quantum Advantage in Boson Sampling Applications*

Professional Services

- **Reviewer for conferences:** QIP (2027, 26, 25, 24, 22, 20), TQC (2025, 24, 23), AQIS (2026, 25), QCE (2023, 21, 20), Asiacrypt 2020
- **Reviewer for journals:** Int. J. Quantum Inf.

Honors & Awards

- PhD Scholarship from Sydney Quantum Academy, 2020-2023
- SUSTech Scholarship for Outstanding Undergraduate, 2016-2017
- University Start-up Scholarship by Shenzhen Municipal Government, 2014-2018
- SUSTech Scholarship for Outstanding Freshman, 2014

Teaching Experience

- TA, Mathematical Methods for Physicists
- TA, Quantum Mechanics I